

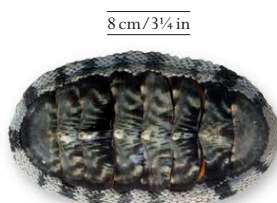
CHITONS

Flattened, rock-hugging chitons are among the most primitive of molluscs. Most are grazers of algae and microbe films, and occur in coastal waters.

A chiton's shell, popularly known as a "coat-of-mail" shell, is made up of eight interlocking plates that are sufficiently flexible to allow the animal to bend when gliding over uneven rocks – or even to roll up when disturbed. Chitons have no eyes or tentacles,

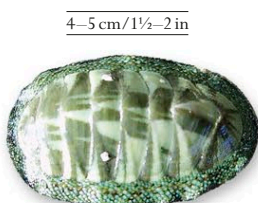
although the shell itself contains cells that enable them to react to light. The plates are fringed by the edge of the mantle, known as the girdle. The mantle forms a fleshy skirt around the chiton. This overhangs grooves running down each side of the animal, which channel water, delivering oxygen to gills that project into the grooves. The radula, or rasping tongue, is coated with minute teeth reinforced with iron and silica, so chitons can graze on the toughest encrusting algae.

PHYLUM	MOLLUSCA
CLASS	POLYPLACOPHORA
ORDERS	2
FAMILIES	20
SPECIES	1,026



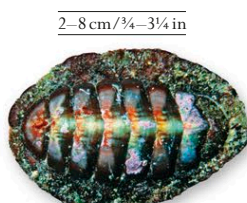
MARbled CHITON
Chiton marmoratus
F: Chitonidae

Like other chitons, the shell plates of this Caribbean mollusc are composed entirely of a chalky mineral called aragonite.



GREEN CHITON
Chiton glaucus
F: Chitonidae

A variable-coloured chiton, this species occurs along the coastlines of New Zealand and Tasmania and, like most chitons, is active at night.



WEST INDIAN FUZZY CHITON
Acanthopleura granulata
F: Chitonidae

This spiky-fringed Caribbean chiton can tolerate exposure to sun, and can live high up in the intertidal zone.



DECKED CHITON
Ischnochiton comptus
F: Ischnochitonidae

Ischnochiton chitons have a spiny or scaly girdle fringe. This is a common intertidal species of W. Pacific shores.



LINED CHITON
Tonicella lineata
F: Ischnochitonidae

This brightly coloured chiton occurs on North Pacific coasts, where it may be camouflaged when feeding on red encrusting algae.



GUMBOOT CHITON
Cryptochiton stelleri
F: Acanthochitonidae

In one chiton family the fleshy girdle overlaps the chain mail plates. In this North Pacific species, the largest chiton, it forms a leathery skin.



BRISTLED CHITON
Mopalia ciliata
F: Mopaliidae

Named for its prominent hairy girdle, this chiton of the Pacific coast of N. America is sometimes found beneath long-moored boats.



HAIRY CHITON
Chaetopleura papilio
F: Chaetopleuridae

A bristly-girdled species with brown-striped chainmail plates, the hairy chiton is found beneath rocks on South African coasts.

TUSK SHELLS

Curious, mud-shovelling tusk shells live in marine sediments and have a tubular curved shell – open at both ends and unlike the shell of any other mollusc.

Tusk shells are common, but are rarely seen because they usually occur far offshore, with the wide base of the hollow tusk buried in mud. The eyeless head and foot reach deeper into the sediment, probing for food with their tentacle-like structures called captacula, and tasting the mud with chemical detectors. When food is found – small invertebrates or detritus – the tentacles bring it to the mouth, where it is ground up by the radula, a rasping tongue typical of molluscs. Tusk shells lack gills; the animal's fleshy mantle encloses a water-filled, oxygen-extracting tube that runs the length of the shell. When levels of oxygen run low, the tusk shell contracts its foot and squirts the stale water out through the top end of the shell. Fresh water enters by the same route.

PHYLUM	MOLLUSCA
CLASS	SCAPHOPODA
ORDERS	2
FAMILIES	13
SPECIES	576

3–4 cm
1¼–1½ in

EUROPEAN TUSK
Antalis dentalis
F: Dentaliidae

A northeast Atlantic species, the European tusk is widespread in sandy areas of coastal waters. In places its empty shells are found in huge numbers.

5–8 cm
2–3¼ in

BEAUTIFUL TUSK
Pictodentalium formosum
F: Dentaliidae

This colourful tropical tusk shell has been recorded in marine sediments in Japan, the Philippines, Australasia, and New Caledonia.

